



Q.1. Give short answers of the following. (15x2=30)

- i. What is the difference between speed and displacement?
- ii. Distinguish between steady and non- steady fluid flow.
- iii. State Torricelli's theorem.
- iv. What do you mean by instantaneous power? Also define horsepower.
- v. The resultant of two vectors of magnitude 3 units and 4 units is $\sqrt{37}$. Calculate the angle between two vectors.
- vi. Calculate the gravitational force between two bodies each of mass 1 kg and separated by a unit distance
- vii. Define term "component of a vector"
- viii. State second condition of equilibrium
- ix. How force and momentum are related to each other? State newton's second law in terms of momentum
- x. Define conservative field. Give at least one example of conservative field
- xi. State work energy principle
- xii. Write down the relationship between linear displacement and angular displacement
- xiii. Compare centripetal force and centrifugal force
- xiv. State Stoke's law. What are its limitations?
- xv. Define terminal velocity

Answers the following questions. (3x10=30)

- Q.2. (a) Define absolute potential energy. Derive relation for absolute potential energy of body of mass 'm' at distance 'r' from the center of earth. (6 Marks)**
- (b) Define and explain moment of inertia? (4 Marks)**
- Q.3. (a) State and explain equation of continuity (6 Marks)**
- (b) A 57kg woman runs up a flight of stairs having a rise of 4.5m in 3.5s. what is the average power must she supply? (4 Marks)**
- Q.4. (a) Discuss interconversion of potential energy and kinetic energy (6 Marks)**
- (b) Derive the relationship between linear and angular acceleration (4 Marks)**